

Namit Seth

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Education

Keshav Mahavidyalaya, Delhi University – B.Sc. (H) Electronics

Expected May 2027

Technical Skills & Concepts

- **Embedded & Hardware:** Microcontrollers (ESP8266, ESP32, ATmega), RTC (DS1302), Sensor Interfacing (MPU6050), PCB Design (KiCad), Simulation (Proteus).
- **Protocols:** I2C, SPI, UART, NEC (IR).
- **Languages & Software:** C/C++ (Embedded), Python (Automation/Regex), JavaScript, SQL, Git.

Projects

ATmega32 Development Board (PCB Design / KiCad) <https://github.com/namit-seth/ATmega32-Development-Board>

- Designed and fabricated a custom ATmega32 development board, from schematic capture to PCB layout.
- Implemented ICSP interfaces for direct microcontroller flashing and designed an op-amp based 2.5V reference circuit driving a bi-directional LED array to visually debug I/O pin states.

Universal IR Hub (ESP8266 / C++)

<https://github.com/namit-seth/Universal-IR-hub>

- Reverse-engineered undocumented NEC protocol (0xF7 series) payloads to develop a universal control system, replacing generic IR libraries.
- Developed custom C++ firmware to process protocol payloads with optimized memory usage, integrating the hardware into the Google Home ecosystem via Sinric Pro.

DS1302 Embedded Library (C++)

<https://github.com/namit-seth/DS1302>

- Replaced high-overhead standard libraries with a lightweight custom RTC driver authored from scratch using hardware datasheet analysis.
- Implemented bit-banged serial communication and direct register-level control, ensuring stable deployment with a minimal footprint in resource-constrained systems.

Standalone Digital Polling System (Microcontroller)

<https://github.com/namit-seth/EVM-V2>

- Resolved signal bouncing and ghost inputs by engineering software-based debouncing logic, eliminating the need for additional physical hardware capacitors.
- Programmed input-handling logic to detect and reject simultaneous multi-button presses, routing validated totals to a 16x2 LCD interface.

Live Multiplayer Treasure Hunt Game (Flask / SQL)

<https://github.com/namit-seth/treasure-hunt>

- Engineered and deployed a real-time web-based treasure hunt for 10 concurrent teams during a departmental festival.
- Diagnosed critical database locks during simultaneous logins, restoring system stability by restructuring transaction management to resolve race conditions.

Experience

Marketing Intern, SecureDApp

Aug 2025 – Sep 2025

- Identified bottlenecks in data extraction pipelines and engineered custom Python scripts utilizing Regex to automate workflows and replace manual scraping.

Full Stack Web Developer Intern, Techplement

Jan 2024 – Feb 2024

- Resolved team code integration challenges by establishing strict Git version control workflows for a collaborative React and Express application, ensuring clean repository merges.

Extracurricular Activities

Secretary, Elektronika, Keshav Mahavidyalaya

2025 – 2026

- Direct the core team in organizing Elexonia, the annual electronics festival, managing technical event logistics and cross-functional student teams.

Executive Member, Student Council, Keshav Mahavidyalaya

2025 – 2026

- Fulfilled administrative duties as an appointed representative, acting as an official signatory to validate student council resolutions.